

Interactive Event: A DIY Pressure Sensitive Chair for Intelligent Tutoring Systems

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This interactive event presents a pressure sensitive chair constructed out of Wii Fit game controller boards. During this event, we will demonstrate how to arrange the boards to detect seat and back pressure, configure a PC to receive a bluetooth datastream from the boards, and modify the power supply of the boards to increase uptime and reliability. We claim that the pressure sensitive chair so constructed is highly suitable for recovering posture information from a student interacting with an ITS.

Research in intelligent tutoring systems is paying increased attention to physiological measures for student modeling. Of particular interest is student affect and engagement which has been measured by multiple groups using posture sensors [1, 2]. Unfortunately, commercially available posture sensors have traditionally been expensive limiting their widespread distribution with ITS.

However, as an alternative Nintendo's Wii Fit game controller can be repurposed as a functional, effective, and cost-efficient posture sensor. Using two Wii Fits, for posterior and back, one can obtain 8 pressure data streams (4 per Fit). Wii Fit boards communicate wirelessly with a PC using bluetooth. Though bluetooth association is a finicky process that is best demonstrated, once associated, the boards remain connected until either the PC is powered down or the boards run out of power. Thus it is ideal to leave the PC on continuously and modify the power supply to accept an AC adapter rather than the native batteries. Though the power requirements are unpublished, a 1000ma adapter soldered to the battery terminals provides good results. Once the boards have been associated, a free software library can be used to interface with the bluetooth stack and retrieve the information sent by the boards, e.g. WiimoteLib (<http://www.codeplex.com/WiimoteLib>). It is simple to log the data streams of a participant for offline analyses, or even process the streams in real time for posture reactive tutor. Provided with an existing chair, a complete system can be constructed for under 200 dollars in less than 1 hour, whereas commercially pressure sensors are available for \$10,000. The cost and robustness of the boards are such that scale up deployment is practical.

References

1. D'Mello, S., Picard, R.W., Graesser, A.: Toward an Affect-Sensitive AutoTutor. *IEEE Intelligent Systems* **22**(4) (2007) 53–61
2. Arroyo, I., Cooper, D.G., Burleson, W., Woolf, B.P., Muldner, K., Christopherson, R.: Emotion sensors go to school. In: *AIED*. (2009) 17–24